

9.4 Orthogonal designs

Section 9.3 showed how much the sums of squares for a term can depend on the other terms. This section asks when they do not. The answer is geometric: after the mean is removed, the terms must span mutually orthogonal subspaces. Balanced experiments achieve this by design and observational data essentially never do, but orthogonality can be manufactured inside a term by a change of basis, as with orthogonal polynomials and orthogonal contrasts.

9.4.1 When the order does not matter

Let the model matrix be $\mathbf{X} = [\mathbf{1}, \mathbf{X}_1, \dots, \mathbf{X}_k]$ with an intercept and k terms. Write $\mathbf{P}_0 = n^{-1}\mathbf{1}\mathbf{1}^\top$, and let

$$\tilde{\mathbf{X}}_j = (\mathbf{I} - \mathbf{P}_0)\mathbf{X}_j, \quad \mathcal{V}_j = \mathcal{C}(\tilde{\mathbf{X}}_j),$$

be the centred columns of term j and the subspace they span. By Lemma 9.2.1 with $\mathbf{X}_1 = \mathbf{1}$, $\mathcal{C}(\mathbf{1}, \mathbf{X}_j) = \text{span}(\mathbf{1}) \oplus \mathcal{V}_j$, so $\text{SS}(\mathbf{X}_j \mid \mathbf{1}) = \|\mathbf{P}_{\mathcal{V}_j}\mathbf{y}\|^2$, the sum of squares of the term fitted on its own after the mean. Call the design **orthogonal** (for these terms) if the subspaces $\mathcal{V}_1, \dots, \mathcal{V}_k$ are mutually orthogonal, that is, if $\tilde{\mathbf{X}}_i^\top \tilde{\mathbf{X}}_j = \mathbf{0}$ for $i \neq j$. For numerical regressors this says that every pair of regressors from different terms has sample correlation zero.

Theorem 9.4.1. Orthogonal designs

For a model $\mathbf{X} = [\mathbf{1}, \mathbf{X}_1, \dots, \mathbf{X}_k]$, the following are equivalent:

- (a) the centred subspaces $\mathcal{V}_1, \dots, \mathcal{V}_k$ are mutually orthogonal;
- (b) $\mathbf{M} = \mathbf{P}_0 + \mathbf{P}_{\mathcal{V}_1} + \dots + \mathbf{P}_{\mathcal{V}_k}$;
- (c) for every j and every $\mathbf{y}$, the marginal and partial sums of squares agree: $\text{SS}(\mathbf{X}_j \mid \mathbf{1}) = \text{SS}(\mathbf{X}_j \mid \mathbf{X}_{(-j)})$;
- (d) for every $\mathbf{y}$, the marginal sums of squares add up to the regression sum of squares: $\sum_j \text{SS}(\mathbf{X}_j \mid \mathbf{1}) = \text{SS}(\mathbf{X}_1, \dots, \mathbf{X}_k \mid \mathbf{1})$.

When they hold, $\text{SS}(\mathbf{X}_j \mid \mathbf{1}, \mathbf{X}_S) = \|\mathbf{P}_{\mathcal{V}_j}\mathbf{y}\|^2$ for every set S of other terms. In particular the sequential sum of squares of $\mathbf{X}_j$ is the same in every order, and it equals the partial and the Type II sums of squares of $\mathbf{X}_j$.

Proof. (a) implies (b) and the final statement. Let S be any set of terms. The subspaces $\text{span}(\mathbf{1})$ and $\mathcal{V}_l$, $l \in S$, are mutually orthogonal, because each $\mathcal{V}_l \subseteq \mathbf{1}^\perp$, and they sum to $\mathcal{C}(\mathbf{1}, \mathbf{X}_S)$, because each $\mathcal{C}(\mathbf{1}, \mathbf{X}_l) = \text{span}(\mathbf{1}) \oplus \mathcal{V}_l$. By Theorem 6.3.7 the projection onto $\mathcal{C}(\mathbf{1}, \mathbf{X}_S)$ is $\mathbf{P}_0 + \sum_{l \in S} \mathbf{P}_{\mathcal{V}_l}$. With S all terms this is (b). With S and $S \cup \{j\}$, the difference of the two projections is $\mathbf{P}_{\mathcal{V}_j}$, which gives $\text{SS}(\mathbf{X}_j \mid \mathbf{1}, \mathbf{X}_S) = \|\mathbf{P}_{\mathcal{V}_j}\mathbf{y}\|^2$. Sequential, partial and Type II sums of squares of $\mathbf{X}_j$ are all of this form.

(b) implies (d): $\text{SS}(\mathbf{X}_1, \dots, \mathbf{X}_k \mid \mathbf{1}) = \mathbf{y}^\top (\mathbf{M} - \mathbf{P}_0) \mathbf{y} = \sum_j \mathbf{y}^\top \mathbf{P}_{\mathcal{V}_j} \mathbf{y}$.

(d) implies (a). The two sides of (d) are the quadratic forms of the symmetric matrices $\sum_j \mathbf{P}_{\mathcal{V}_j}$ and $\mathbf{M} - \mathbf{P}_0$. Two symmetric matrices with the same quadratic form are equal, because $2\mathbf{u}^\top \mathbf{A} \mathbf{v} = (\mathbf{u} + \mathbf{v})^\top \mathbf{A} (\mathbf{u} + \mathbf{v}) - \mathbf{u}^\top \mathbf{A} \mathbf{u} - \mathbf{v}^\top \mathbf{A} \mathbf{v}$. So $\sum_j \mathbf{P}_{\mathcal{V}_j} = \mathbf{M} - \mathbf{P}_0$, which is idempotent (Theorem 6.5.1). By the algebraic form of Cochran's theorem (Theorem 4.5.1), projections whose sum is idempotent satisfy $\mathbf{P}_{\mathcal{V}_i} \mathbf{P}_{\mathcal{V}_j} = \mathbf{0}$ for $i \neq j$, which is (a).

(a) implies (c) is part of the final statement.

(c) *implies* (a). As before, equality of the quadratic forms gives $\mathbf{P}_{\mathcal{V}_j} = \mathbf{M} - \mathbf{M}_{(-j)}$. The right-hand side projects onto $\mathcal{C}(\mathbf{X}) \ominus \mathcal{C}(\mathbf{X}_{(-j)})$ (Theorem 6.5.1), which is orthogonal to $\mathcal{C}(\mathbf{X}_{(-j)})$. The space $\mathcal{C}(\mathbf{X}_{(-j)})$ contains $\mathcal{V}_i$ for every $i \neq j$, since $\tilde{\mathbf{X}}_i = \mathbf{X}_i - \mathbf{P}_0 \mathbf{X}_i$ has columns in $\mathcal{C}(\mathbf{1}, \mathbf{X}_i)$. So $\mathcal{V}_j \perp \mathcal{V}_i$. ■

Idea. Orthogonality is a property of the *centred* subspaces. The raw columns of two indicator variables are never orthogonal to the intercept, and a regressor and its square are rarely orthogonal. What matters is whether the parts of the terms that vary around their means point in perpendicular directions.

Orthogonality also stabilizes the coefficients.

Corollary 9.4.2. Coefficients in orthogonal designs

Suppose $\mathbf{X} = [\mathbf{1}, \mathbf{X}_1, \dots, \mathbf{X}_k]$ has full column rank and satisfies the conditions of Theorem 9.4.1. Then the least squares coefficients of the columns of $\mathbf{X}_j$ are the same in every model that contains $\mathbf{1}$ and $\mathbf{X}_j$, whichever other terms it contains. Only the intercept changes.

Proof. In the model with terms $S \ni j$, the fitted vector is $\mathbf{P}_0 \mathbf{y} + \sum_{l \in S} \mathbf{P}_{\mathcal{V}_l} \mathbf{y}$ by the proof of the theorem, and $\mathbf{P}_{\mathcal{V}_l} \mathbf{y} = \tilde{\mathbf{X}}_l \mathbf{b}_l$ with $\mathbf{b}_l = (\tilde{\mathbf{X}}_l^\top \tilde{\mathbf{X}}_l)^{-1} \tilde{\mathbf{X}}_l^\top \mathbf{y}$, which does not depend on S . Since $\tilde{\mathbf{X}}_l \mathbf{b}_l = \mathbf{X}_l \mathbf{b}_l - \mathbf{P}_0 \mathbf{X}_l \mathbf{b}_l$, the fitted vector equals $\sum_{l \in S} \mathbf{X}_l \mathbf{b}_l$ plus a multiple of $\mathbf{1}$. The coefficients are unique by full rank, so the coefficient vector of $\mathbf{X}_j$ is $\mathbf{b}_j$ in every such model. ■

With uncorrelated regressors, dropping one leaves the others' estimates as they were, and each regressor has an order-free sum of squares. This is one reason factorial experiments are laid out on orthogonal arrays (Chapters 16 and 18).

9.4.2 Balanced and proportional layouts

For two factors the orthogonality condition reduces to a condition on the cell counts.

Theorem 9.4.3. Proportional frequencies

In a two-way layout with factor A at a levels, factor B at b levels, cell counts n_{ij} , and all row totals $n_{i\cdot}$ and column totals $n_{\cdot j}$ positive, the centred subspaces of A and B are orthogonal iff

$$n_{ij} = \frac{n_{i\cdot} n_{\cdot j}}{n} \quad \text{for all } i, j. \quad (9.4.1)$$

Proof. A vector in the centred subspace of A takes a value α_i on every observation in row i , with $\sum_i n_{i\cdot} \alpha_i = 0$. A vector in that of B takes the value β_j in column j , with $\sum_j n_{\cdot j} \beta_j = 0$. Their inner product is $\sum_{i,j} n_{ij} \alpha_i \beta_j = \boldsymbol{\alpha}^\top \mathbf{N} \boldsymbol{\beta}$, with $\mathbf{N} = (n_{ij})$. Write $\mathbf{r} = (n_{1\cdot}, \dots, n_{a\cdot})^\top$ and

$\mathbf{c} = (n_{.1}, \dots, n_{.b})^\top$. The subspaces are orthogonal iff $\boldsymbol{\alpha}^\top \mathbf{N} \boldsymbol{\beta} = 0$ whenever $\boldsymbol{\alpha}^\top \mathbf{r} = 0$ and $\mathbf{c}^\top \boldsymbol{\beta} = 0$.

If Eq. 9.4.1 holds, $\mathbf{N} = \mathbf{r} \mathbf{c}^\top / n$ and $\boldsymbol{\alpha}^\top \mathbf{N} \boldsymbol{\beta} = (\boldsymbol{\alpha}^\top \mathbf{r})(\mathbf{c}^\top \boldsymbol{\beta}) / n = 0$. Conversely, let $\mathbf{E} = \mathbf{N} - \mathbf{r} \mathbf{c}^\top / n$. Its row sums are $\mathbf{r} - \mathbf{r}(\mathbf{c}^\top \mathbf{1}) / n = \mathbf{0}$ and its column sums are $\mathbf{c}^\top - (\mathbf{1}^\top \mathbf{r}) \mathbf{c}^\top / n = \mathbf{0}^\top$, since $\mathbf{1}^\top \mathbf{r} = \mathbf{c}^\top \mathbf{1} = n$. So $\boldsymbol{\alpha}^\top \mathbf{E} \boldsymbol{\beta}$ does not change when $\boldsymbol{\alpha}$ or $\boldsymbol{\beta}$ is shifted by a multiple of $\mathbf{1}$. Given any $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$, shift them to $\boldsymbol{\alpha}' = \boldsymbol{\alpha} - (\boldsymbol{\alpha}^\top \mathbf{r} / n) \mathbf{1}$ and $\boldsymbol{\beta}' = \boldsymbol{\beta} - (\mathbf{c}^\top \boldsymbol{\beta} / n) \mathbf{1}$, which satisfy $\boldsymbol{\alpha}'^\top \mathbf{r} = 0$ and $\mathbf{c}^\top \boldsymbol{\beta}' = 0$. Then $\boldsymbol{\alpha}^\top \mathbf{E} \boldsymbol{\beta} = \boldsymbol{\alpha}'^\top \mathbf{E} \boldsymbol{\beta}' = \boldsymbol{\alpha}'^\top \mathbf{N} \boldsymbol{\beta}' - 0 = 0$ by assumption. As $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ are arbitrary, $\mathbf{E} = \mathbf{0}$. ■

Equal replication, $n_{ij} = m$, is the common case (Example (Additive two-factor model, one observation per cell), Exercise (C1)). If the interaction subspace is defined as the orthogonal complement of the additive model within the cell-constant vectors, then under Eq. 9.4.1 the subspaces for the mean, A , B and AB form a decomposition whatever the order.

Orthogonality settles Type I against Type II, but not Type III. Under proportional frequencies the Type I and Type II sums of squares of A coincide and address equality of row means weighted by the column proportions $n_{.j}/n$ (Exercise (B2)), while Type III weights the columns equally. They agree with equal replication or without interaction.

Example. Firms and years

Grunfeld's investment panel (Example (Grunfeld's investment data)) records 11 firms in each of 20 years, one observation per firm and year. As a two-way layout with factors firm and year, it has $n_{ij} = 1$, so Eq. 9.4.1 holds, and the additive model's sums of squares for the logarithm of investment are order-free:

Source	df	Sum of squares
firm	10	473.482
year	19	26.749
residual	190	14.667
corrected total	219	514.898

Firms differ enormously in the scale of their investment, and years matter much less. Adding the logarithm of market value as a covariate destroys the orthogonality, because value varies with both firm and year. The sum of squares for firm then depends on what else is in the model: it is 90.701 after year and value, and 88.443 after value alone. Most of the variation between firms is variation in size, which value already measures. This is the setting of analysis of covariance (Chapter 18, Theorem 18.1.1).

```
import numpy as np
import pandas as pd
import statsmodels.api as sm

def proj(*blocks):
    Z = np.column_stack(blocks)
    U, s, _ = np.linalg.svd(Z, full_matrices=False)
    U = U[:, s > 1e-10 * s[0]]
    return U @ U.T
```

```

g = sm.datasets.grunfeld.load_pandas().data
y = np.log(g["invest"].to_numpy())
n = len(y)
one = np.ones((n, 1))
F = pd.get_dummies(g["firm"]).to_numpy(float)          # 11 firms
T = pd.get_dummies(g["year"]).to_numpy(float)          # 20 years
P0, PF, PT, PFT = proj(one), proj(one, F), proj(one, T), proj(one, F, T)

print("centred spaces orthogonal:", np.abs((PF - P0) @ (PT - P0)).max() < 1e-10)
print(f"firm:  first {y @ (PF - P0) @ y:8.3f}    after year {y @ (PFT - PT) @ y:8.3f}"
      ")
print(f"year:  first {y @ (PT - P0) @ y:8.3f}    after firm {y @ (PFT - PF) @ y:8.3f}"
      ")

```

[Run this code online](#)

```

v = np.log(g["value"].to_numpy())[:, None]              # log market value
PFv, PTv, PFTv = proj(one, F, v), proj(one, T, v), proj(one, F, T, v)
print(f"with log value: firm after year, value {y @ (PFTv - PTv) @ y:8.3f}"
      f"    firm after value {y @ (PFv - proj(one, v)) @ y:8.3f}")

```

[Run this code online](#)

9.4.3 Orthogonal polynomials

A polynomial regression in a single variable x , with distinct values $x_1, \dots, x_n$, has a built-in chain of models,

$$\mathcal{P}_0 \subseteq \mathcal{P}_1 \subseteq \dots \subseteq \mathcal{P}_d, \quad \mathcal{P}_k = \text{span}(\mathbf{1}, \mathbf{x}, \mathbf{x}^2, \dots, \mathbf{x}^k),$$

where $\mathbf{x}^k$ is the vector with entries x_i^k . The decomposition of interest is that of the chain (Theorem 9.1.1(e)). Its k th subspace, $\mathcal{P}_{k-1}^\perp \cap \mathcal{P}_k$, is a line. Applying Gram–Schmidt (Proposition 6.2.2) to $\mathbf{1}, \mathbf{x}, \mathbf{x}^2, \dots$ produces unit vectors $\mathbf{q}_0, \mathbf{q}_1, \dots, \mathbf{q}_d$ with $\text{span}(\mathbf{q}_0, \dots, \mathbf{q}_k) = \mathcal{P}_k$ for every k . So $\mathbf{q}_k$ spans that line. The vectors $\mathbf{q}_k$, evaluated at the design points, are the **orthogonal polynomials** of the design. Three consequences follow at once.

1. The sequential sum of squares for degree k , the drop in residual sum of squares from degree $k-1$ to degree k , is $(\mathbf{q}_k^\top \mathbf{y})^2$, with one degree of freedom.
2. The columns $\mathbf{q}_1, \dots, \mathbf{q}_d$ are orthogonal and centred, so as separate terms they form an orthogonal design (Theorem 9.4.1), and their sums of squares do not depend on the order of entry.
3. The coefficient of $\mathbf{q}_k$ is $\mathbf{q}_k^\top \mathbf{y}$ in every fit that includes it (Corollary 9.4.2). Raising the degree never changes the lower coefficients.

For equally spaced x the orthogonal polynomials have integer multiples that are widely tabulated. For three points they are $(-1, 0, 1)$ and $(1, -2, 1)$. For four points they are $(-3, -1, 1, 3)$, $(1, -1, -1, 1)$ and $(-1, 3, -3, 1)$. They serve as contrasts for a quantitative factor with equally spaced levels, splitting the treatment sum of squares into linear, quadratic and higher components. Their numerical merits and the recurrence that generates them belong to Chapter 42 (Theorem 42.2.2).

Example. A trend in the Nile

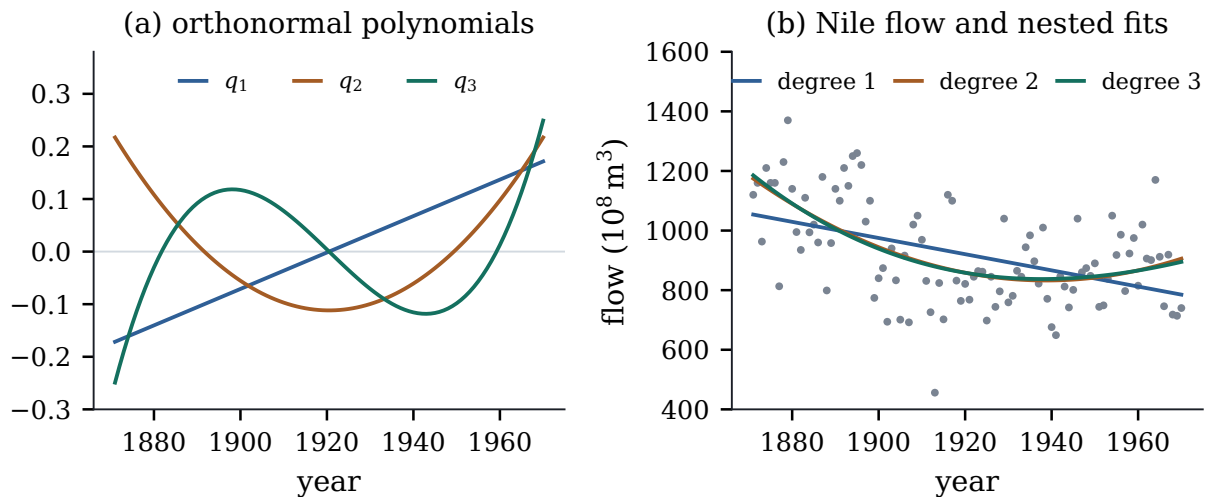


Figure 9.4.1: (a) The orthonormal polynomials $\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3$ for the years 1871–1970. (b) The Nile flow with the nested least squares fits of degrees 1, 2 and 3. The quadratic and cubic fits nearly coincide, because the cubic sum of squares is tiny.

The annual flow of the Nile at Aswan was recorded for the 100 years 1871–1970. How much of its variation is a smooth trend? The listing builds the orthonormal polynomials $\mathbf{q}_0, \dots, \mathbf{q}_4$ for the years by a QR factorization of the matrix of powers, and reads off the sequential sums of squares (in units of $(10^8 \text{ m}^3)^2$):

Degree	1	2	3	4
Sequential sum of squares	613893	309415	1894	134142

The corrected total is 2835157. The linear term accounts for a fraction 0.217 of it, and the first three terms together for 0.326. Against the residual mean square of the cubic fit, 19895 on 96 degrees of freedom, the quadratic component has $F = 15.55$ and the cubic only 0.10. The quartic sum of squares is large again, a sign that the pattern is not a gentle polynomial at all. The flow dropped abruptly around the end of the nineteenth century (Cobb 1978), and a low-degree polynomial can only approximate a step (Figure 9.4.1).

With the raw powers of the centred and scaled year, the sums of squares depend on the order: the linear column entered after the quadratic and cubic ones receives 74751 instead of 613893. The raw linear coefficient is -134.4 in the linear and quadratic fits, but -117.3 in the cubic and quartic ones. It survives the quadratic term because the years are symmetric about their mean, which makes odd and even centred powers orthogonal (Exercise (C1)), but not the cubic term. The coefficient of $\mathbf{q}_1$ is -783.5 in every fit.

```
import numpy as np
import statsmodels.api as sm
nile = sm.datasets.nile.load_pandas().data
year = nile["year"].to_numpy()
y = nile["volume"].to_numpy()

t = year - year.mean()
V = np.vander(t / t.max(), 5, increasing=True)
Q, R = np.linalg.qr(V)
Q = Q * np.sign(np.diag(R))
    with tk                                # centring only helps rounding
                                           # 1, t, t2, t3, t4 (scaled)
                                           # Gram-Schmidt in matrix form
                                           # make each qk agree in sign
```

```

coef = Q.T @ y                                     # coordinates of y on q_0, ...,
    q_4
seq_ss = coef[1:] ** 2                             # sequential SS: linear, ...,
    quartic
for k, s in enumerate(seq_ss, start=1):
    print(f"degree {k}: sequential SS = {s:10.0f}")
print(f"residual after degree 4: {y @ y - np.sum(coef ** 2):10.0f}")

```

[Run this code online](#)

9.4.4 Single-degree-of-freedom decompositions

Orthogonal polynomials split a sum of squares into one-dimensional pieces along directions chosen for their meaning. The same can be done for any hypothesis, by choosing single estimable functions whose estimates are uncorrelated.

Theorem 9.4.4. Orthogonal single-degree-of-freedom sums of squares

Let $\lambda_1, \dots, \lambda_s$ be nonzero vectors with $\lambda_k^\top \beta$ estimable, so that $\lambda_k = \mathbf{X}^\top \rho_k$, and let $\mathbf{G}$ be a generalized inverse of $\mathbf{X}^\top \mathbf{X}$.

- (a) The sum of squares for $\lambda_k^\top \beta = 0$ is

$$SS_k = \frac{(\lambda_k^\top \hat{\beta})^2}{\lambda_k^\top \mathbf{G} \lambda_k} = \|\mathbf{P}_{\mathbf{v}_k} \mathbf{y}\|^2, \quad \mathbf{v}_k = \mathbf{M} \rho_k,$$

the squared length of the projection of $\mathbf{y}$ onto the line spanned by $\mathbf{v}_k$.

- (b) $\mathbf{v}_h^\top \mathbf{v}_k = \lambda_h^\top \mathbf{G} \lambda_k$. If $\lambda_h^\top \mathbf{G} \lambda_k = 0$ for all $h \neq k$, the lines are mutually orthogonal, and $\sum_k SS_k$ is the sum of squares for the joint hypothesis $\lambda_1^\top \beta = \dots = \lambda_s^\top \beta = 0$, on s degrees of freedom.
- (c) Conversely, every orthonormal basis $\mathbf{u}_1, \dots, \mathbf{u}_s$ of a subspace of $\mathcal{C}(\mathbf{X})$ arises in this way, with $\lambda_k = \mathbf{X}^\top \mathbf{u}_k$.

Proof.

- (a) is Theorem 9.2.4 with $q = 1$ and $\mathbf{d} = 0$: the test space is spanned by $\mathbf{M} \rho_k = \mathbf{v}_k$, and the projection of $\mathbf{y}$ onto it has squared length $(\mathbf{v}_k^\top \mathbf{y})^2 / \mathbf{v}_k^\top \mathbf{v}_k$, where $\mathbf{v}_k^\top \mathbf{y} = \rho_k^\top \mathbf{M} \mathbf{y} = \lambda_k^\top \hat{\beta}$ and $\mathbf{v}_k^\top \mathbf{v}_k = \lambda_k^\top \mathbf{G} \lambda_k$. (b) $\mathbf{v}_h^\top \mathbf{v}_k = \rho_h^\top \mathbf{M} \rho_k = \rho_h^\top \mathbf{X} \mathbf{G} \mathbf{X}^\top \rho_k = \lambda_h^\top \mathbf{G} \lambda_k$ by Theorem 6.3.6. Orthogonal nonzero vectors are independent, so $\Lambda = [\lambda_1, \dots, \lambda_s]$ has rank s (because $\mathbf{X}^\top \mathbf{V} = \Lambda$ with $\mathbf{V} = [\mathbf{v}_1, \dots, \mathbf{v}_s]$, and $\mathbf{X}^\top$ is one-to-one on $\mathcal{C}(\mathbf{X})$). The test space of the joint hypothesis is $\mathcal{C}(\mathbf{M} \mathbf{T}) = \mathcal{C}(\mathbf{V})$, with $\mathbf{T} = [\rho_1, \dots, \rho_s]$, by Theorem 9.2.4(b), and the projection onto a span of orthogonal lines is the sum of the projections onto the lines (Theorem 6.3.7). (c) With $\rho_k = \mathbf{u}_k$, $\mathbf{v}_k = \mathbf{M} \mathbf{u}_k = \mathbf{u}_k$, and $\lambda_k^\top \mathbf{G} \lambda_k = \mathbf{u}_k^\top \mathbf{u}_k$. ■

In the one-way layout with group sizes $n_1, \dots, n_g$, take the generalized inverse of Exercise (B3). A contrast $\sum_i c_i \alpha_i$ with $\sum_i c_i = 0$ (Definition 8.7.1) is estimated by $\sum_i c_i \bar{y}_i$, and $\lambda^\top \mathbf{G} \lambda = \sum_i c_i^2 / n_i$.

So two contrasts $\mathbf{c}$ and $\mathbf{c}'$ give orthogonal sums of squares iff

$$\sum_{i=1}^g \frac{c_i c'_i}{n_i} = 0,$$

and $g - 1$ such contrasts split the between-groups sum of squares into single degrees of freedom. With equal group sizes this is ordinary orthogonality, $\sum_i c_i c'_i = 0$; with unequal sizes, contrasts that look orthogonal need not be.

Example. Splitting the education sum of squares

Consider the 944 self-placements of Example (Party, education and political self-placement) by education alone. The group sizes are 313, 277 and 354, and the group means are 4.476, 4.357 and 4.167. The between-groups sum of squares is 16.306 on two degrees of freedom. The contrast of college against graduate, $\mathbf{c}_2 = (0, 1, -1)$, has sum of squares 5.653. The contrast of high school against the other two, weighted by their sizes, $\mathbf{c}_1 = (1, -0.439, -0.561)$, is orthogonal to $\mathbf{c}_2$ in the sense above, and has sum of squares 10.652. The two add up to the between-groups sum of squares, up to rounding. The more obvious contrast $(1, -\frac{1}{2}, -\frac{1}{2})$, high school against the plain average of the others, has sum of squares 9.534, and together with $\mathbf{c}_2$ it gives 15.187: with unequal group sizes it is not orthogonal to $\mathbf{c}_2$.

```
import numpy as np
import pandas as pd
import statsmodels.api as sm
d = sm.datasets.anes96.load_pandas().data
educ = pd.cut(d["educ"], [0, 3.5, 5.5, 7], labels=["HS", "College", "Graduate"])
y = d["selfLR"].to_numpy()

counts = educ.value_counts(sort=False).to_numpy().astype(float)
means = d.groupby(educ, observed=True)["selfLR"].mean().to_numpy()
grand = y.mean()
ss_treat = np.sum(counts * (means - grand) ** 2)          # SS(educ | 1), 2 df

def contrast_ss(c):
    """(c' ybar)^2 / sum(c_i^2 / n_i): the SS for the hypothesis c' mu = 0."""
    return (c @ means) ** 2 / np.sum(c ** 2 / counts)

def orthogonal(c, h):
    return np.isclose(np.sum(c * h / counts), 0.0)

n2, n3 = counts[1], counts[2]
c1 = np.array([1.0, -n2 / (n2 + n3), -n3 / (n2 + n3)]) # HS against the rest,
# weighted
c2 = np.array([0.0, 1.0, -1.0])                       # College against Graduate
c1_plain = np.array([1.0, -0.5, -0.5])                 # HS against the plain
# average
print("orthogonal:", orthogonal(c1, c2), orthogonal(c1_plain, c2))
print(f"treatment SS {ss_treat:.3f} = {contrast_ss(c1):.3f} + {contrast_ss(c2):.3f}")
print(f"with the plain contrast: {contrast_ss(c1_plain):.3f} + {contrast_ss(c2):.3f}")
```

[Run this code online](#)

Single-degree-of-freedom sums of squares reappear in [Chapter 13](#), where many of them are tested

at once, and in [Chapter 15](#), where orthogonal contrasts organize the one-way analysis of variance (Theorem 15.4.1).

9.4.5 Exercises

9.4.5.1 A. Check your understanding

Exercise. A1

With $x = 1, 2, 3, 4$ and $\mathbf{y} = (3, 5, 4, 8)^\top$, use the orthogonal polynomial contrasts for four equally spaced points to compute the linear, quadratic and cubic sums of squares, and check that they add up to the corrected total sum of squares.

Exercise. A2

Which of the 2×2 count tables $\begin{pmatrix} 2 & 4 \\ 3 & 6 \end{pmatrix}$, $\begin{pmatrix} 2 & 4 \\ 4 & 2 \end{pmatrix}$ and $\begin{pmatrix} 1 & 3 \\ 3 & 9 \end{pmatrix}$ give orthogonal centred factor spaces? For the one that does not, compute the inner product of the two centred indicator vectors of the first levels.

Exercise. A3

Show that if the regressors $\mathbf{x}_1, \dots, \mathbf{x}_k$ of a model with an intercept are pairwise uncorrelated, then $R^2 = \sum_j r_{yj}^2$, where r_{yj} is the sample correlation of $\mathbf{y}$ and $\mathbf{x}_j$.

9.4.5.2 B. Practice

Exercise. B1

In a two-way layout with interaction and $n_{ij} = m \geq 1$ for all cells, show that the Type I sums of squares of A in either order, and its Type II and Type III sums of squares, are all equal to $bm \sum_i (\bar{y}_{i..} - \bar{y}_{...})^2$.

Solution. Equal counts satisfy [Eq. 9.4.1](#), so by [Theorem 9.4.1](#) the Type I sums of squares of A in both orders and its Type II sum of squares equal $\text{SS}(A \mid \mu) = \|(\mathbf{P}_A - \mathbf{P}_0)\mathbf{y}\|^2$. Each row has bm observations, and $\mathbf{P}_A - \mathbf{P}_0$ replaces an observation in row i by $\bar{y}_{i..} - \bar{y}_{...}$, which gives the formula. For Type III, the hypothesis of [Theorem 9.3.2\(b\)](#) is that of (a), because with equal counts weighted and unweighted row means coincide. A sum of squares for a hypothesis depends only on the hypothesis (it is $\text{SSE}_0 - \text{SSE}$), so the Type III sum of squares equals $\text{SS}(A \mid \mu)$ too.

Exercise. B2

Take counts $n_{11} = 1, n_{12} = 2, n_{21} = 2, n_{22} = 4$. Check [Eq. 9.4.1](#). Find cell means for which the Type III hypothesis for A holds but the Type I (and Type II) hypothesis fails.

Exercise. B3

In a one-way layout with g groups of equal size m , show that the Helmert contrasts $\mathbf{c}_k = (1, \dots, 1, -(k-1), 0, \dots, 0)$, with $k-1$ leading ones, $k = 2, \dots, g$, are mutually orthogonal, and write the between-groups sum of squares as a sum of $g-1$ single-degree-of-freedom sums of squares. What goes wrong with unequal group sizes?

9.4.5.3 C. Going deeper

Exercise. C1

Suppose the design points are symmetric about their mean: after sorting, $x_i - \bar{x} = -(x_{n+1-i} - \bar{x})$. Show that every odd power of the centred variable is orthogonal to every even power (including the zeroth), that $\mathbf{q}_k$ is an even or odd vector according to the parity of k , and explain the pattern of raw coefficients in [Example \(A trend in the Nile\)](#).

Solution. Let $\mathbf{t}$ have entries $t_i = x_i - \bar{x}$ and let $\mathbf{J}$ be the reversal permutation, so that $\mathbf{J}\mathbf{t} = -\mathbf{t}$. A vector $\mathbf{u}$ is even if $\mathbf{J}\mathbf{u} = \mathbf{u}$ and odd if $\mathbf{J}\mathbf{u} = -\mathbf{u}$. Powers $\mathbf{t}^k$ have the parity of k , and an even vector $\mathbf{u}$ and an odd vector $\mathbf{v}$ are orthogonal because $\mathbf{u}^\top \mathbf{v} = (\mathbf{J}\mathbf{u})^\top (\mathbf{J}\mathbf{v}) = -\mathbf{u}^\top \mathbf{v}$. Gram-Schmidt on $\mathbf{1}, \mathbf{t}, \mathbf{t}^2, \dots$ subtracts from $\mathbf{t}^k$ only its projections on the earlier $\mathbf{q}_j$ of the same parity, by induction, so $\mathbf{q}_k$ has the parity of k . Hence the column $\mathbf{t}$ is orthogonal to the column $\mathbf{t}^2$ and to $\mathbf{1}$, and adding the quadratic term does not change the linear coefficient ([Corollary 9.4.2](#), applied to the terms $\mathbf{t}$ and $\mathbf{t}^2 - \bar{\mathbf{t}}^2 \mathbf{1}$). The cubic column is odd and not orthogonal to $\mathbf{t}$, so it changes the linear coefficient. The quartic column is even. By Theorem 9.2.2(b), adding it changes the cubic fit's coefficients by $-\mathbf{L}\hat{\gamma}$ with $\mathbf{L} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{t}^4$, where $\mathbf{X} = [\mathbf{1}, \mathbf{t}, \mathbf{t}^2, \mathbf{t}^3]$. Inner products between odd and even columns vanish, so $\mathbf{X}^\top \mathbf{X}$ and its inverse do not link the odd and even columns, and the entries of $\mathbf{X}^\top \mathbf{t}^4$ for the odd columns are zero. Hence the rows of $\mathbf{L}$ for $\mathbf{t}$ and $\mathbf{t}^3$ vanish, and the linear coefficient is unchanged.